

Realistic Multi-Cloud Dust-Scattering Halos and Maximum-Likelihood Localization of Gravitational-Wave X-ray Counterparts

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Abstract

Nederlander & Paerels (2020) proposed that a prompt X-ray flash from a gravitational-wave (GW) source will scatter off Galactic dust and produce an expanding ring-shaped halo whose photons arrive hours to a day after the event, and whose geometric center pinpoints the source. Their treatment used two idealized dust geometries and asserted, without an explicit algorithm, that the halo center can be located to arcsecond precision. Here I (i) build a Monte-Carlo forward model of scattering halos for realistic line-of-sight dust distributions consisting of multiple discrete screens, which produce a set of concentric, differently expanding rings; (ii) implement an explicit maximum-likelihood centroiding estimator for the source position; and (iii) quantify the achievable localization precision with simulations. The recovered localization error scales as $N^{-1/2}$ with the number of detected halo photons N , reaching $\approx 4''$ at $N = 30$ and $\approx 0.6''$ at $N = 1000$, and saturates an analytic Cramér–Rao bound $2\sigma_r/\sqrt{N}$ set by the ring width σ_r measured from the simulated halos. The precision is robust to a realistic soft X-ray background and degrades by only a few percent when the halo comprises several nested rings from dust at different distances. I further show that the source can be localized *without* any prior knowledge of the dust distribution, using only the concentricity of the rings; that the same analysis tomographically recovers the distances to the intervening dust screens (60, 150, 400 pc recovered as 60, 151, 403 pc); that both properties survive realistic many-cloud Galactic sightlines (median localization error $1.3''$ over an ensemble of 50 random sightlines); and that the halo’s $\sqrt{t}$ expansion between epochs measures the dust distances independently and rejects spurious detections. A short detectability analysis shows the ring is a $> 5\sigma$ feature for a few tens of photons and is reliably localized for a few hundred, collapsing a $100\text{--}1000\text{ deg}^2$ GW skymap to a few square arcseconds. This confirms and sharpens the central claim of Nederlander & Paerels (2020): a detected scattering halo localizes its GW source finely enough to identify a unique host galaxy, and does so even when the foreground dust is unknown.

1. Introduction

Gravitational-wave observatories localize compact-binary mergers only coarsely — typically tens to hundreds of square degrees — which makes prompt identification of an electromagnetic (EM) counterpart difficult, as the follow-up campaign for GW170817 illustrated even in a favorable case (Abbott et al. 2017; Evans

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et al. 2017). For binary black-hole mergers, which are not expected to produce the neutron-star merger kilonova, any EM signal is likely to be a brief, faint flash that all-sky monitors may miss entirely. Determining a precise sky position is essential both for identifying the host galaxy (and hence the redshift, breaking distance–inclination degeneracies) and for triggering follow-up.

Nederlander & Paerels (2020, hereafter *NP20*) pointed out that if such an event emits X-rays, those photons will scatter off interstellar dust in our own Galaxy and generate a scattering halo — a phenomenon predicted by Overbeck (1965), since observed around bright Galactic X-ray sources (e.g., Mauche & Gorenstein 1986; Predehl et al. 2000; Heinz et al. 2015) and as expanding rings after gamma-ray bursts (e.g., Vaughan et al. 2004; Tiengo & Mereghetti 2006), and extended theoretically to extragalactic and intergalactic dust (Corrales 2015). Because scattered photons travel a slightly longer geometric path, they arrive with a delay of hours to a day, providing a “reprieve” in which the counterpart can still be found even if the prompt flash was missed. Crucially, the halo is a set of rings centered on the source direction, so its centroid recovers the source position with far greater precision than the GW localization.

NP20 developed the feasibility of this idea using two deliberately extreme dust geometries — a single thin screen at 100 pc and dust distributed uniformly to 100 pc — and estimated that the halo center could be located to of order $1''$ – $10''$. They did not, however, model the clumpy, multi-cloud dust distribution that real sightlines present, nor did they construct and test an explicit localization algorithm. This paper addresses both gaps. Section 2 sets out the halo physics and a Monte-Carlo forward model that admits an arbitrary set of dust screens, together with a maximum-likelihood centroiding estimator. Section 3 presents simulated halos, a quantitative characterization of localization precision as a function of photon count, background, and dust geometry, a self-calibrating estimator that requires no prior dust model, a demonstration that the halo tomographically recovers the foreground dust distances, a realistic many-cloud Galactic sightline, and the halo’s time evolution. Section 4 develops the detectability and a concrete search strategy. Section 5 discusses implications for host-galaxy identification and the limitations of the present treatment.

2. Method

2.1. Halo geometry and time delay

Consider a source at an effectively infinite (extragalactic) distance whose prompt X-rays illuminate a thin dust screen a distance d from the observer, oriented perpendicular to the line of sight. A photon scattered through a small angle θ is observed at angular offset θ from the source direction and, having travelled a slightly longer path, arrives with a delay

$$t = \frac{d}{2c} \theta^2,$$

so that at a fixed observation time t after the flash the screen is seen as a ring of angular radius

$$\theta_{\text{ring}}(t, d) = \sqrt{\frac{2ct}{d}}.$$

For $d = 100$ pc this gives a ring radius of $7.0'$ at a delay of 6 hr, reproducing the value quoted by NP20. A nearer screen produces a larger, faster-expanding ring and a more distant screen a smaller, slower one; several screens along the same sightline therefore yield a set of concentric rings sharing a common center at the

source position. During a finite exposure each ring has a small radial width set by the range of delay times sampled, and is further broadened by the telescope point-spread function (PSF).

2.2. Scattering cross section

In the Rayleigh–Gans regime the differential scattering cross section is approximately Gaussian in the scattering angle (Mauche & Gorenstein 1986; Draine 2011),

$$\frac{d\sigma}{d\Omega} \propto \exp\left(-\frac{\theta^2}{2\theta_0^2}\right), \quad \theta_0 = 10.4 \left(\frac{E}{\text{keV}}\right)^{-1} \left(\frac{a}{0.1 \mu\text{m}}\right)^{-1} \text{ arcmin},$$

where E is the photon energy and a the characteristic grain radius. For $E = 1 \text{ keV}$ and $a = 0.1 \mu\text{m}$, $\theta_0 = 10.4'$. This factor weights how many photons populate each ring: rings whose radius greatly exceeds θ_0 are strongly suppressed. In the forward model each simulated photon is accepted with probability proportional to the cross section evaluated at the ring radius from which it originates. The Rayleigh–Gans approximation is known to overestimate the scattering below $\approx 1 \text{ keV}$ relative to the exact Mie solution (Smith & Dwek 1998), so the fiducial 1 keV case sits at the edge of its formal validity; because this modifies only the brightness weighting of the rings and not their geometry (Section 5), the localization results below are insensitive to it.

2.3. Forward model

The simulator takes a source position, a list of dust screens (each specified by distance, relative scattering optical depth, and grain size), a photon energy, an exposure window $[t_0, t_0 + t_{\text{exp}}]$, a PSF width, and a uniform background surface density. For each halo photon it draws a parent screen (weighted by optical depth), a scattering time within the exposure window, and a random azimuth; computes the ring radius from Eq. (2); accepts the photon against the differential cross section at that radius (Section 2.2); and applies a Gaussian PSF blur. Background photons are drawn uniformly across the field. The output is a detected photon list — exactly the observable a real X-ray imager would deliver. Throughout I adopt a Gaussian PSF of dispersion $\sigma_{\text{PSF}} = 0.15' = 9''$ (half-power diameter $\approx 0.35'$, comparable to the $0.3'$ of *Swift*/XRT, the instrument NP20 considered) and a $40' \times 40'$ field.

The framework accepts an arbitrary set of screens, so it can represent both the idealized cases used for validation and physically motivated sightlines. For the latter (Section 3.7) I generate cloud distances from an exponential disk-like profile and column densities (hence relative optical depths) from a log-normal distribution, and I draw grain radii per photon from a scattering-weighted size distribution: an MRN number law $n(a) \propto a^{-3.5}$ (Mathis, Rumpl, & Nordsieck 1977) combined with an X-ray scattering cross section $\sigma_{\text{scat}} \propto a^4$ yields an effective $p(a) \propto a^{+0.5}$ over $0.05\text{--}0.25 \mu\text{m}$, so that larger grains dominate the scattered flux; each photon is then accepted against the full differential cross section, whose $1/\theta_0^2 \propto a^2$ normalization further concentrates the large-grain contribution at small angles. Because the ring *radius* is fixed by geometry, the grain-size spread affects only the relative brightness of the rings, not their positions.

2.4. Maximum-likelihood localization

Given the detected photon list and the known ring template (fixed by the assumed screen distances and the exposure window), the only free parameters are the source coordinates (x_c, y_c) . I model the sky intensity as a mixture of the PSF-smear ring template $s(r)$, evaluated at each photon’s radius r from the trial center, and a flat background b with fractional weight f , and maximize the log-likelihood

$$\ln \mathcal{L}(x_c, y_c) = \sum_j \ln[(1-f)s(r_j) + fb], \quad r_j = \sqrt{(x_j - x_c)^2 + (y_j - y_c)^2}.$$

The radial template is precomputed once on a fine grid so that evaluating the likelihood at any trial center reduces to a single interpolation per photon; the best center is found by a coarse grid search followed by two stages of local refinement. This estimator uses only the ring geometry, not photon energies, and makes no assumption of a single ring — it applies unchanged to the multi-screen case.

2.5. The analytic localization limit

The precision achievable by *any* unbiased estimator is set by the Fisher information and does not depend on the particular algorithm. Consider a single ring of mean angular radius θ_{ring} and radial width σ_r (from the exposure duration and PSF), with photons distributed uniformly in azimuth. The probability density of one photon at position (x, y) relative to the true center is

$$p(x, y) = \frac{1}{2\pi r} \frac{1}{\sqrt{2\pi} \sigma_r} \exp\left[-\frac{(r - \theta_{\text{ring}})^2}{2\sigma_r^2}\right], \quad r = \sqrt{x^2 + y^2}.$$

Because p depends on position only through r , and $\partial r / \partial x_c = -\cos \phi$, the Fisher information for the center coordinate x_c per photon is $I_{xx} = \langle (\partial_{x_c} \ln p)^2 \rangle = \langle \cos^2 \phi \rangle \langle (\partial_r \ln p)^2 \rangle$. With $\langle \cos^2 \phi \rangle = 1/2$ and, for a thin ring ($\sigma_r \ll \theta_{\text{ring}}$), $\langle (\partial_r \ln p)^2 \rangle \simeq 1/\sigma_r^2$, this gives $I_{xx} = I_{yy} = 1/(2\sigma_r^2)$ per photon and $I_{xy} = 0$. For N photons the Cramér–Rao bound on the two-dimensional RMS localization error is therefore

$$\Delta \theta_{\text{loc}} = \sqrt{\text{Var}(x_c) + \text{Var}(y_c)} \geq \frac{2\sigma_r}{\sqrt{N}}.$$

The error is set by the ring *width* divided by $\sqrt{N}$, independent of the ring radius. Section 3.3 shows that the maximum-likelihood estimator saturates this bound.

3. Results

3.1. A single-screen halo

Figure 1 shows a simulated halo from a single screen at 100 pc observed in a 1300 s exposure beginning 6 hr after the flash, with 60 halo photons and a background of 0.07 photons arcmin⁻² (about half the diffuse soft X-ray background rate NP20 adopt for this exposure; Section 3.4 shows the precision is insensitive to this choice over an order of magnitude). The scattered photons trace a clean ring at 7'; the likelihood map peaks sharply at the true source position, which the estimator recovers to within an arcsecond.

3.2. Realistic multi-cloud halos

Real sightlines contain several discrete dust clouds. Figure 2 shows a halo from three screens at 60, 150, and 400 pc, which appear as concentric rings at 9.2', 5.8', and 3.6' respectively at a delay of ≈ 6 hr. This nested-ring morphology is a qualitatively new prediction relative to the single- and uniform-screen cases of NP20, and it carries extra information: the set of ring radii at a given time encodes the line-of-sight dust distribution, while all rings remain centered on the source. The likelihood surface (right panel) shows the

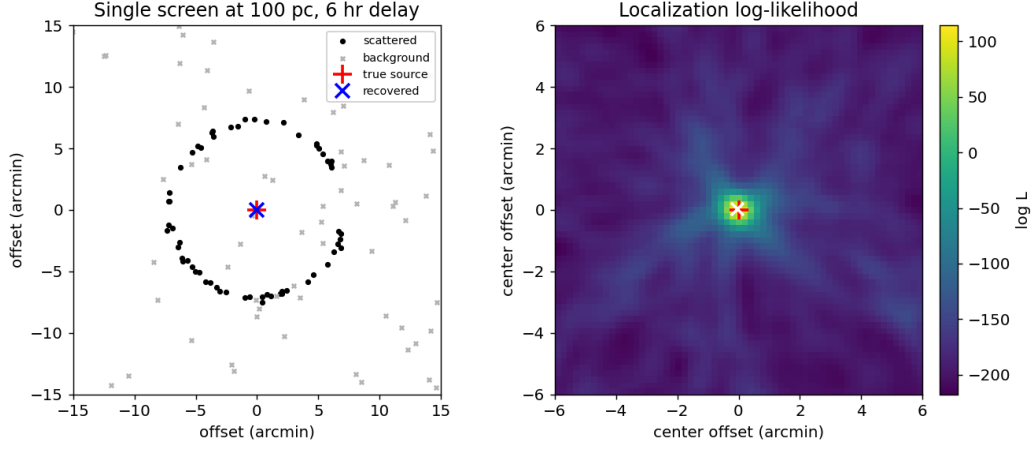


Figure 1: Left: simulated single-screen halo (black points) plus background (grey crosses); the true source (red +) and the maximum-likelihood center (blue x) coincide. Right: the localization log-likelihood over trial centers, sharply peaked at the source.

sharp central peak plus weaker concentric secondary maxima where pairs of rings can be mutually aligned; these are well separated from the true center and do not bias the recovered position.

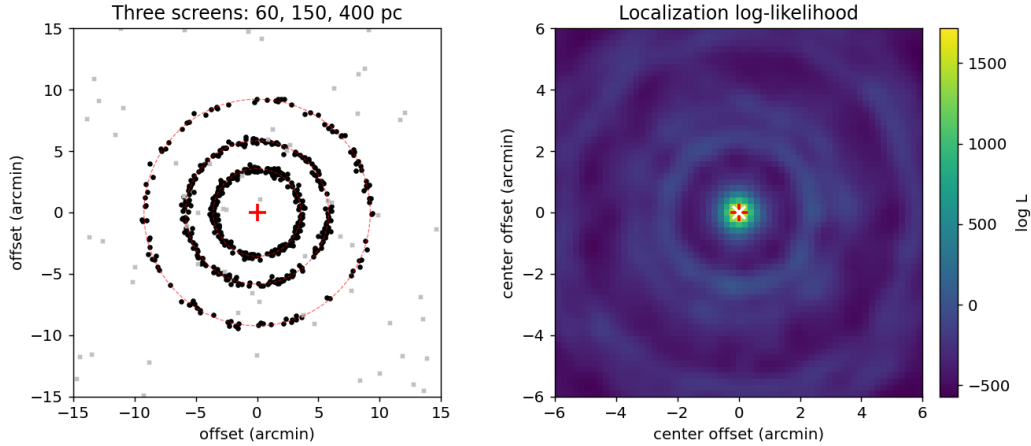


Figure 2: Left: simulated three-screen halo; dashed red circles mark the predicted ring radii for dust at 60, 150, and 400 pc. Right: the localization log-likelihood, with a dominant central peak and weaker concentric aliases.

3.3. Localization precision

To characterize the achievable precision I generated 120 independent realizations at each photon count and measured the RMS offset of the recovered center from the truth. Figure 3 and Table 1 give the results. The localization error falls from $\approx 4''$ at $N = 30$ to $\approx 0.6''$ at $N = 1000$ detected halo photons and tracks the Cramér–Rao bound of Section 2.5 almost exactly: measuring the ring width from the simulations gives $\sigma_r = 9.7''$, and the prediction $2\sigma_r/\sqrt{N}$ passes through the background-free single-screen points to within $\lesssim 10\%$ over the entire range (e.g., $1.23''$ predicted versus $1.21''$ measured at $N = 250$). The bound is derived for one ring of width σ_r observed without background, so the other two configurations are expected to sit slightly above it, and do: the agreement is within $\lesssim 10\%$ for $N \gtrsim 60$ but degrades at the sparsest count

(18% and 30% at $N = 30$ for the background and three-screen cases), where occasional near-failures of the estimator inflate the RMS. The maximum-likelihood estimator is thus statistically efficient — it extracts essentially all the localization information in the halo. It is also unbiased and well-calibrated: over 220 realizations at $N = 200$ the mean recovered offset is $0.09''$ (against a $\sim 1''$ scatter), the RMS matches the Cramér–Rao prediction to 8%, and 62% of trials fall within the nominal 68% error region — consistent with near-Gaussian errors only marginally broader than the ideal bound. A realistic background degrades the precision only mildly. The three-screen halo localizes essentially as well as the single ring at the same total photon count, because every ring constrains the same center: the measured three-screen errors sit $\approx 6\%$ above the single-screen-with-background values, a 2σ difference in the Monte Carlo.

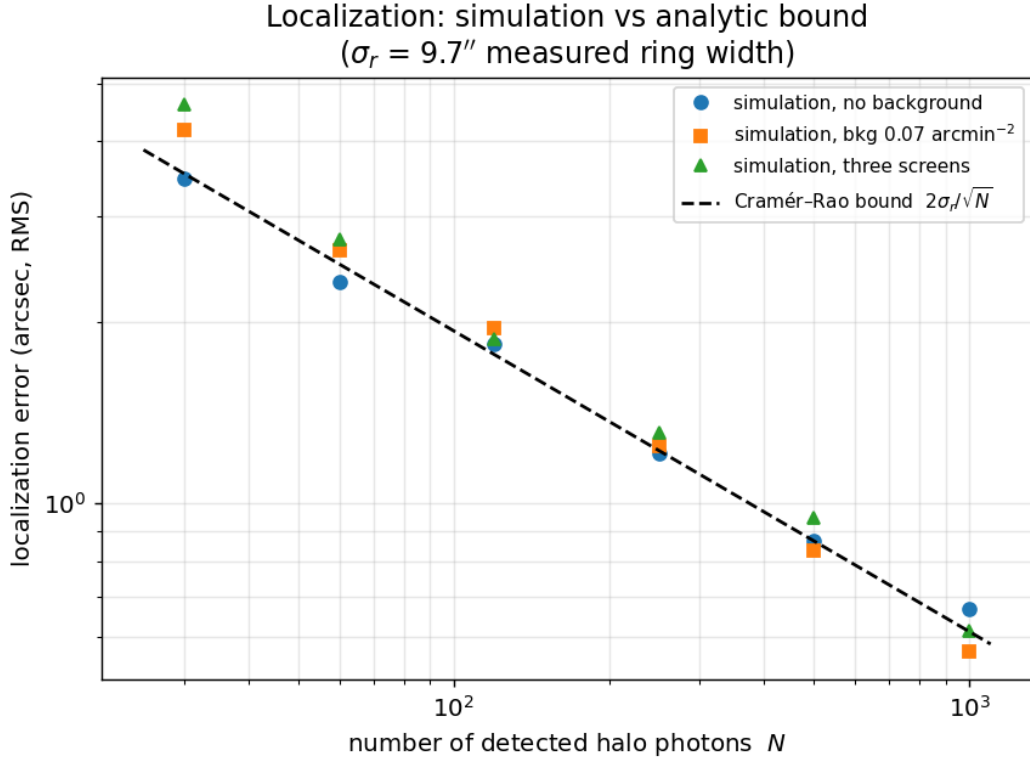


Figure 3: RMS localization error versus number of detected halo photons for three configurations (points), compared with the analytic Cramér–Rao bound $2\sigma_r/\sqrt{N}$ (dashed) using the measured ring width $\sigma_r = 9.7''$.

Table 1: RMS localization error (arcsec) versus detected halo photon count.

N (halo photons)	Single screen, no background	Single screen, bkg 0.07 arcmin^{-2}	Three screens, bkg 0.07 arcmin^{-2}
30	3.5	4.2	4.6
60	2.3	2.6	2.8
120	1.8	2.0	1.9
250	1.2	1.3	1.3
500	0.87	0.84	0.95
1000	0.67	0.57	0.61

3.4. Robustness to background

Figure 4 fixes the halo at 300 photons and varies the background surface density over an order of magnitude (135 trials per point). The localization error is nearly flat, rising only from $\approx 1.1''$ in the background-free limit to $\approx 1.2''$ at $0.6 \text{ photons arcmin}^{-2}$ — several times the reference soft X-ray background. The ring is a strong geometric constraint: even when background photons outnumber halo photons across the field, the concentric structure fixes the center at the arcsecond level.

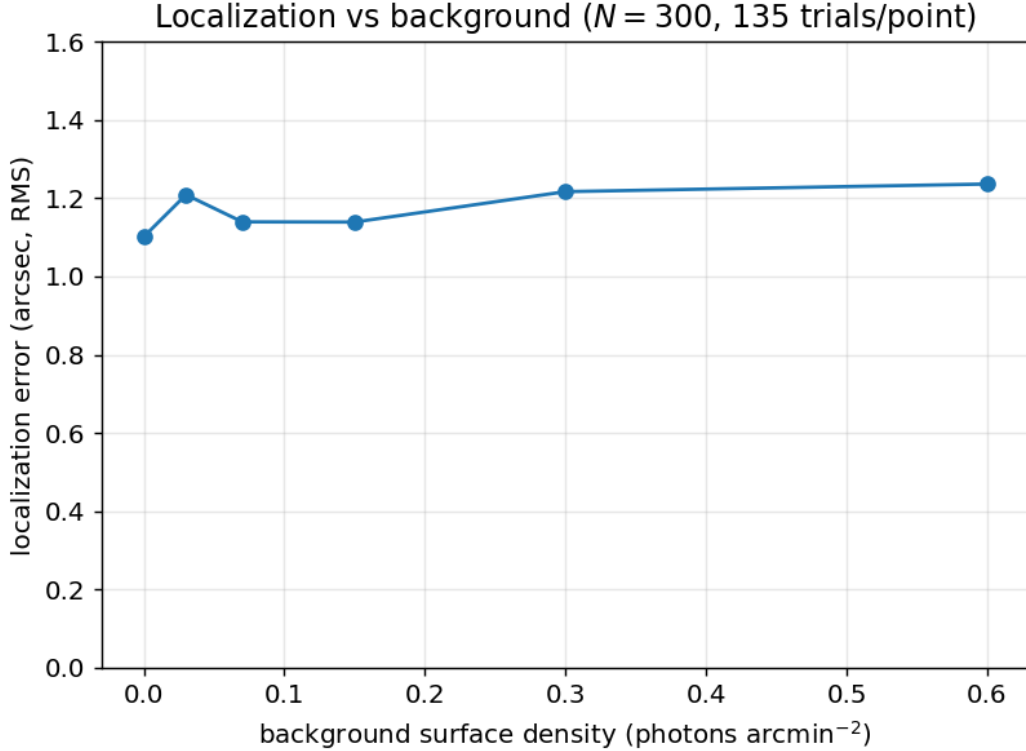


Figure 4: RMS localization error versus background surface density at fixed $N = 300$ halo photons.

3.5. Localizing without knowing the dust distribution

The likelihood of Section 2.4 assumes the dust-screen distances are known, so that the ring radii can be predicted. In practice a sightline’s dust distribution may be uncertain. I therefore also implemented a *self-calibrating* estimator that uses no distance information at all: it exploits only the fact that the rings are concentric about the source. For a trial center, each photon’s radius is computed and a smoothed radial density is built from the data themselves (with the geometric rdr area factor divided out so that a uniform background does not mimic a ring); the score is the summed log-density of the photons. The correct center is the one at which the radii pile up most sharply into rings. This estimator needs neither the screen distances nor the number of screens.

Figure 5 compares the two estimators for the three-screen halo, over 120 independent realizations at each photon count. Once more than a few dozen photons are detected, the self-calibrating estimator tracks the distance-informed one to within a factor of 1.4–2.0, and both follow the $N^{-1/2}$ scaling. The main penalty for discarding distance information appears at the sparsest photon counts: at $N = 30$ (about ten photons per ring)

the self-calibrating estimator fails catastrophically — locking onto a spurious center more than $30''$ from the truth — in 22.5% of trials, whereas at $N = 60$ no such failure occurs in 120 trials (a rate below $\approx 3\%$ at 95% confidence). The practical implication is that the source can be localized directly from the halo image alone, without an external dust model, provided the halo is more than marginally detected.

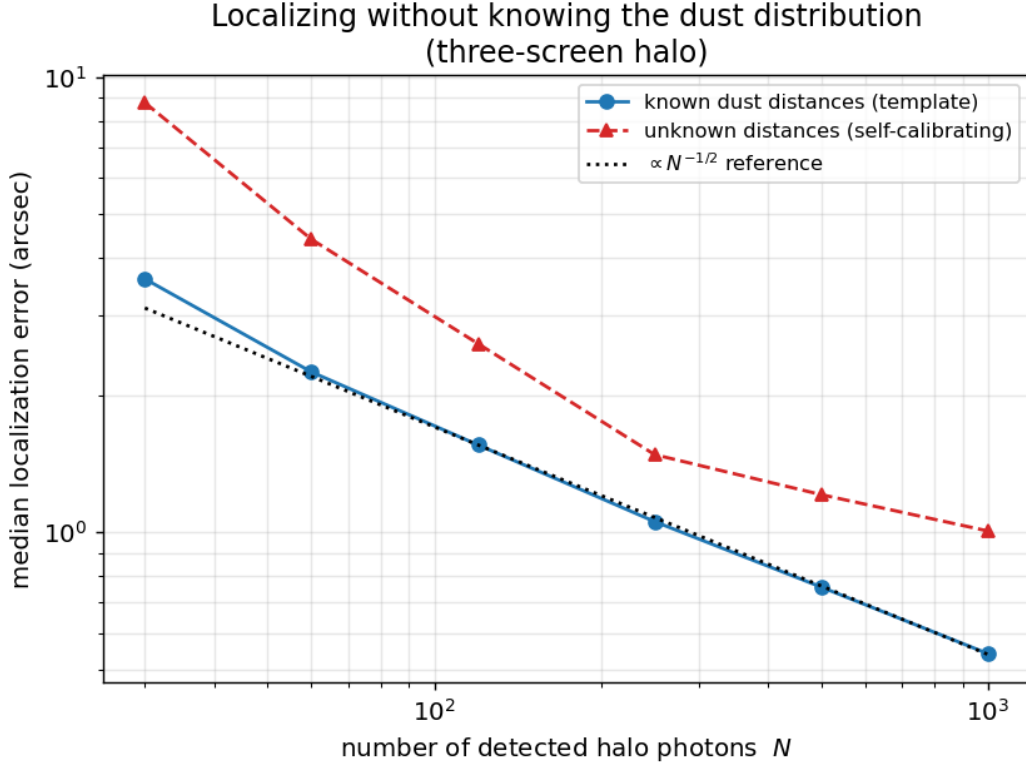


Figure 5: Median localization error versus photon count for the three-screen halo, comparing the distance-informed template estimator with the self-calibrating estimator that uses no dust-distance information.

3.6. Halo tomography: recovering the dust distribution

A corollary of self-calibration is that, once the center is found, the radii of the recovered rings measure the line-of-sight dust distribution. Inverting the geometry, a ring of angular radius θ seen at delay t corresponds to a screen at distance $d = 2ct/\theta^2$. Figure 6 shows this for the three-screen halo: the radial profile about the recovered center resolves three clean peaks, and converting their radii recovers screen distances of 60, 151, and 403 pc, against true values of 60, 150, and 400 pc. A scattering halo thus does double duty — it localizes the source *and* tomographically maps the intervening Galactic dust, a diagnostic that a single steady Galactic X-ray source cannot provide because it lacks a known flash time.

3.7. A realistic Galactic sightline

The three-screen case is a controlled test; a real sightline contains many clouds spread over a range of distances. Figure 7 shows a halo for a sightline drawn from the disk-and-log-normal model of Section 2.3 — eight clouds between 65 and 1021 pc, with a scattering-weighted grain-size distribution — imaged with 900 halo photons. The result is a superposition of many partly overlapping rings that merge into a structured,

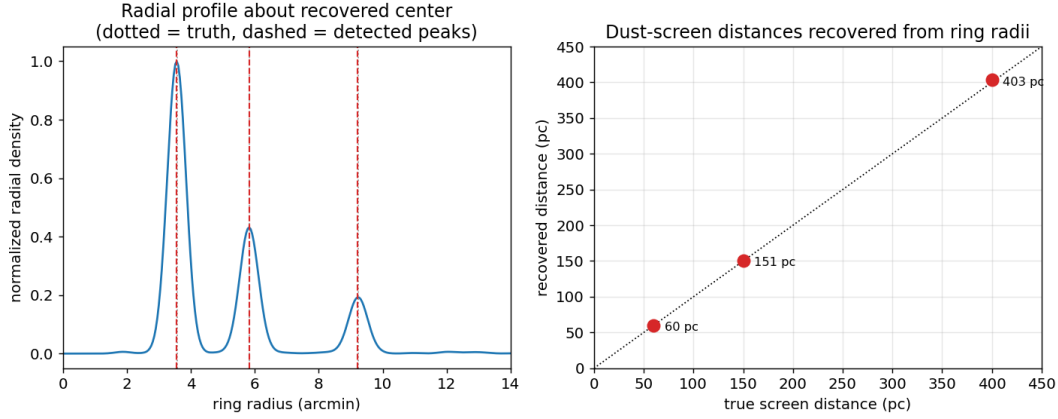


Figure 6: Halo tomography for the three-screen case. Left: radial density about the recovered center (dotted, truth; dashed, detected peaks). Right: dust-screen distances recovered from the ring radii versus the true distances (dotted line is equality).

filled halo rather than a set of clean annuli. Two conclusions survive this complexity. First, localization survives: the self-calibrating estimator recovers the source to $2.2''$ with no knowledge of the eight cloud distances, because concentricity is a property of *all* the rings simultaneously. Second, tomography degrades gracefully: the dominant clouds are still recovered as distinct radial peaks (at 463 and 195 pc, against a true 437–489 pc pair and a cloud at exactly 195 pc), while closely spaced clouds blend into single peaks and the outermost rings — strongly suppressed by the scattering cross section — fall below the peak-detection threshold. Nor is this example a lucky draw: over an ensemble of 50 random sightlines generated from the same model (5–20 clouds each, same photon budget), the self-calibrating estimator achieves a median error of $1.3''$, a 90th-percentile error of $2.6''$, and a worst case of $3.4''$ — every sightline in the ensemble is localized to better than $5''$. A scattering halo therefore remains a clean localizer, and a coarse dust probe, even for a messy line of sight.

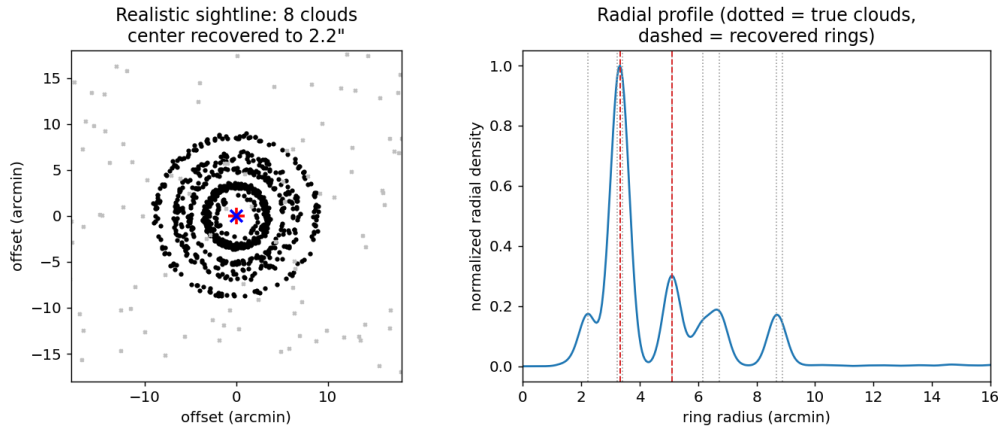


Figure 7: A realistic sightline of eight dust clouds (65–1021 pc). Left: the resulting multi-ring halo; the source is recovered to $2.2''$ (red +, true; blue x, recovered). Right: the radial profile about the recovered center, with true clouds (dotted) and recovered rings (dashed).

3.8. Time evolution and false-positive rejection

A scattering halo is a transient: because $\theta_{\text{ring}} = \sqrt{2ct/d}$, each ring expands as the square root of the elapsed time. Figure 8 (left) images a single 100 pc screen at two epochs; the ring grows from $5.0'$ at 3 hr to $8.6'$ at 9 hr, a ratio of exactly $\sqrt{3}$. Figure 8 (right) tracks the recovered ring radii of the three-screen halo across six epochs from 2 to 12 hr; each ring follows its predicted $\sqrt{2ct/d}$ track, so the *rate* of expansion independently measures each cloud's distance. This time dependence is also a powerful discriminant against spurious halos: real scattered light expands about a fixed center between exposures, whereas background structure and instrumental artifacts do not. Requiring a coherent $\sqrt{t}$ expansion across two or more epochs therefore rejects false positives that a single image cannot.

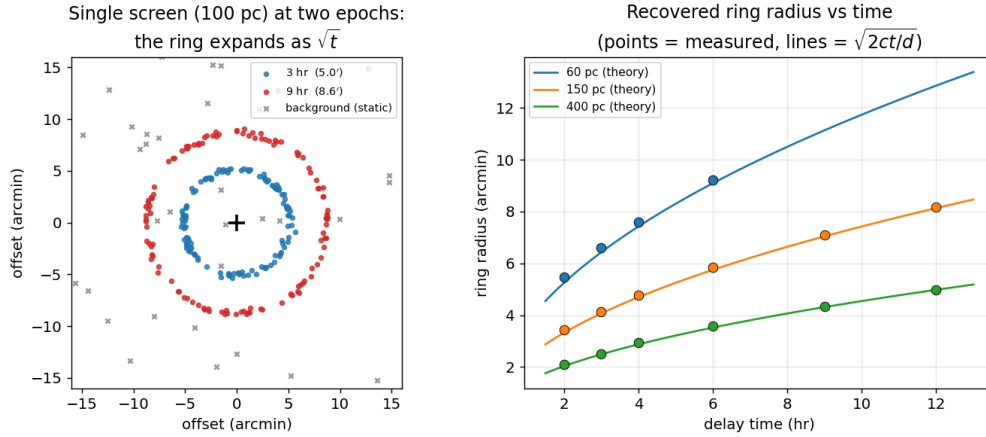


Figure 8: Left: a single 100 pc screen imaged at 3 hr and 9 hr; the ring expands as $\sqrt{t}$ while background points (grey) stay fixed. Right: recovered ring radii versus delay time for three screens (points), with the $\sqrt{2ct/d}$ prediction for each distance (curves).

4. Detectability and search strategy

Two questions determine whether the method is usable in practice: is the halo detectable above background, and, once detected, is it localized? Figure 9 (left) gives the detection significance of the ring for a single 100 pc screen, whose annulus (radius $7'$, width $\sim 1'$) subtends $\approx 44 \text{ arcmin}^2$. Because the ring occupies a small area, the enclosed background is modest and the significance $S/\sqrt{S+B}$ exceeds 5σ for as few as ≈ 30 photons in the annulus, only weakly dependent on the background level over the range considered. Figure 9 (right) shows the operational localization-success fraction — the fraction of trials localized to better than $5''$ by the self-calibrating estimator — as a function of detected photons. For a reference background of 0.07 arcmin^{-2} the success fraction reaches $\approx 90\%$ by ~ 120 photons and unity by ~ 300 ; a heavier background of 0.3 arcmin^{-2} shifts the curve to higher photon counts but the halo is still reliably localized once a few hundred photons are collected.

These numbers imply a concrete search strategy. Following an unlocalized GW trigger, a wide-field soft X-ray imager (e.g., a lobster-eye instrument of the Einstein Probe/WXT type, or an eROSITA-class survey telescope; Predehl 2017) should tile the (hundreds of square degree) skymap in short repeated exposures over the first day, seeking an extended, centrally symmetric excess. A candidate is confirmed by re-imaging: a genuine halo expands as $\sqrt{t}$ about a fixed center (Section 3.8), which rejects static false positives. Once confirmed, the center is obtained from the self-calibrating estimator with no external dust model, and the ring

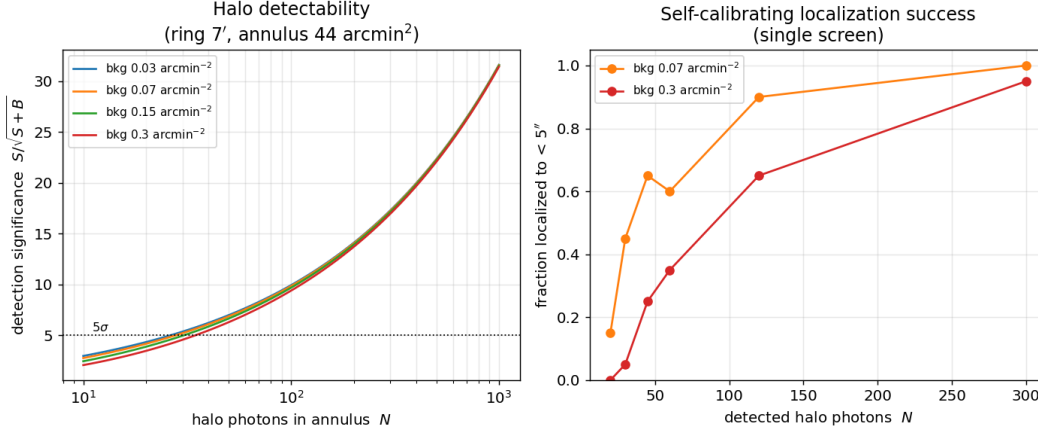


Figure 9: Left: halo detection significance $S/\sqrt{S+B}$ versus photons in the ring annulus, for four background levels (5σ marked). Right: fraction of trials localized to better than $5''$ by the self-calibrating estimator, versus detected photon count, for two backgrounds.

radii yield the foreground dust distances as a by-product. The pay-off is a dramatic shrinkage of the search region: locating the source to $\sim 1''$ collapses a $100\text{--}1000\text{ deg}^2$ GW skymap to a few square arcseconds, a reduction in area of $\sim 4 \times 10^8\text{--}4 \times 10^9$, which is what makes a unique host-galaxy identification possible.

4.1. Matched-filter detection and the trials factor

The annulus estimate above is deliberately conservative because it discards the ring’s shape. The optimal linear detector is a matched filter: cross-correlating the image with the zero-mean radial ring template $T(r)$ gives a statistic $D(c) = \sum_j T(|x_j - c|)$ whose maximum over trial centers (on a $0.2'$ grid, finer than the ring width, covering the part of the field in which the ring fits entirely) is the field’s detection statistic. Under background alone the maxima cluster tightly (Figure 10, left), whereas even a faint halo of $N = 60$ photons lands far into the tail. Fitting the background maxima with a Gumbel distribution (the appropriate extreme-value form for a maximum) and requiring a global false-alarm probability of 2.9×10^{-7} (5σ) sets the detection threshold; the required photon count is defined as that at which the *mean* halo statistic reaches the threshold, i.e. a 50% detection efficiency. Because a survey must tile many fields, the threshold rises with the number of independent fields M — but only logarithmically. Figure 10 (right) shows that the halo photons needed for a global 5σ detection grow from ≈ 17 for a single field to just ≈ 23 across 10^4 tiled fields: the ring is such a distinctive template that the trials penalty of an all-sky search is mild. This assumes the ring radius — set by the unknown dust distance — is known; a real search runs a bank of templates spanning plausible distances, which multiplies the effective trials by the modest number of independent radii and leaves the required count at a few tens of photons. The matched filter is therefore somewhat more sensitive than the annulus estimate of the previous paragraph, and confirms that detection, like localization, is limited by photons rather than by background or survey size.

4.2. Expected performance for a fiducial event

The localization can be tied to a physical event using the photon budget of NP20. For a source at 400 Mpc emitting a fraction $\varepsilon = 10^{-5}\varepsilon_{-5}$ of the GW energy in soft X-rays, scattered by a 100 pc dust path and collected by a telescope of order 100 cm^2 effective area (the *Swift*/XRT class NP20 considered), NP20 estimate $\approx 4500\varepsilon_{-5}$ scattered photons in total; since the Gaussian cross section puts only 60% of them inside

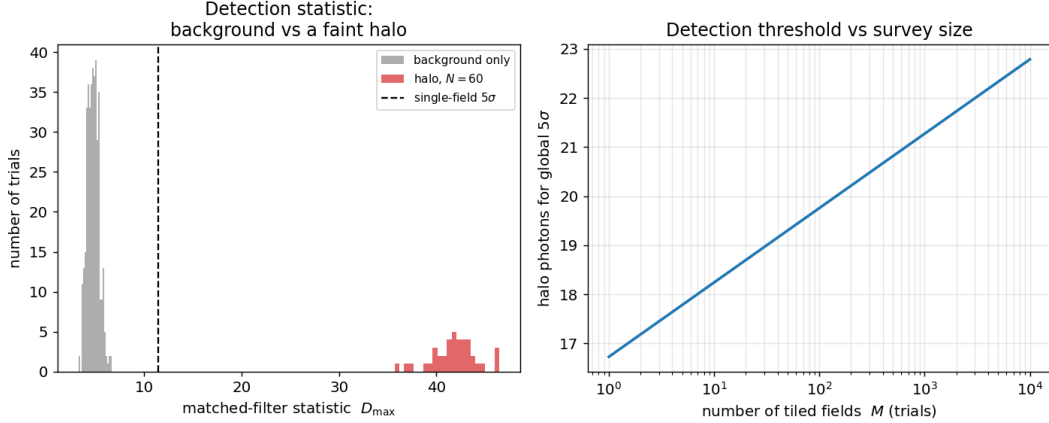


Figure 10: Left: distribution of the matched-filter detection statistic $D_{\max}$ for background-only fields (grey) and for fields containing a faint $N = 60$ halo (red), with the single-field 5σ threshold (dashed). Right: halo photons required for a global 5σ detection versus the number of tiled fields M ; the trials penalty grows only logarithmically.

the ring radius reached at 24 hr (NP20 quote $t_{1/2} \approx 18$ hr for half the light), a day-long campaign can collect $N \approx 2700 \varepsilon_{-5}$. Exploiting this budget requires combining exposures across epochs, which the estimator accommodates naturally, since the rings at every epoch share the same center (Section 3.8). Combining this with the localization law $\Delta\theta_{\text{loc}} = 2\sigma_r/\sqrt{N}$ (Section 2.5) gives the expected precision as a function of the — highly uncertain — X-ray efficiency. Figure 11 shows the result, with the simulated points confirming the scaling. Even for an efficiency as low as $\varepsilon_{-5} = 0.01$ (about 27 photons) the source is localized to $\approx 4''$, and for $\varepsilon_{-5} \gtrsim 0.03$ to better than $5''$; across the whole plausible range the error lies far below the $\sim 35''$ mean separation between candidate host galaxies. The conclusion is encouraging: whenever the halo is detectable at all, it is localizable to the host-galaxy level. The binding uncertainty is astrophysical — whether black-hole mergers emit detectable X-rays — not the localization method.

5. Discussion

The practical question is whether the halo centroid pins down a *unique* host galaxy. At the design sensitivity of LIGO/Virgo, $30M_{\odot}$ black-hole binaries are detectable to ≈ 1.5 Gpc; for a galaxy density of $\approx 10^{-2} \text{ Mpc}^{-3}$ the mean separation between candidate hosts in the relevant volume is tens of arcseconds (Chen & Holz 2016; NP20). The arcsecond-level localization found here is therefore more than sufficient to isolate a single host in the large majority of cases, confirming the central claim of NP20 with an explicit algorithm and a quantified error budget.

Two themes run through these results. The first is robustness: the localization is essentially unchanged by a realistic eight-cloud sightline (Section 3.7), by soft X-ray background (Section 3.4), and by discarding all prior knowledge of the dust (Section 3.5), because concentricity is a global property of the whole halo rather than of any one ring. The second is that the foreground dust — normally a nuisance — becomes information: the ring radii measure the screen distances (Section 3.6), and their $\sqrt{t}$ expansion both confirms the transient nature of the signal and rejects static false positives (Section 3.8), underpinning the search strategy of Section 4.

Two points of physics are worth clarifying. The ring radius at a fixed delay, $\theta = \sqrt{2ct/d}$, follows from the geometric path-length difference and is therefore independent of photon energy; energy enters only through

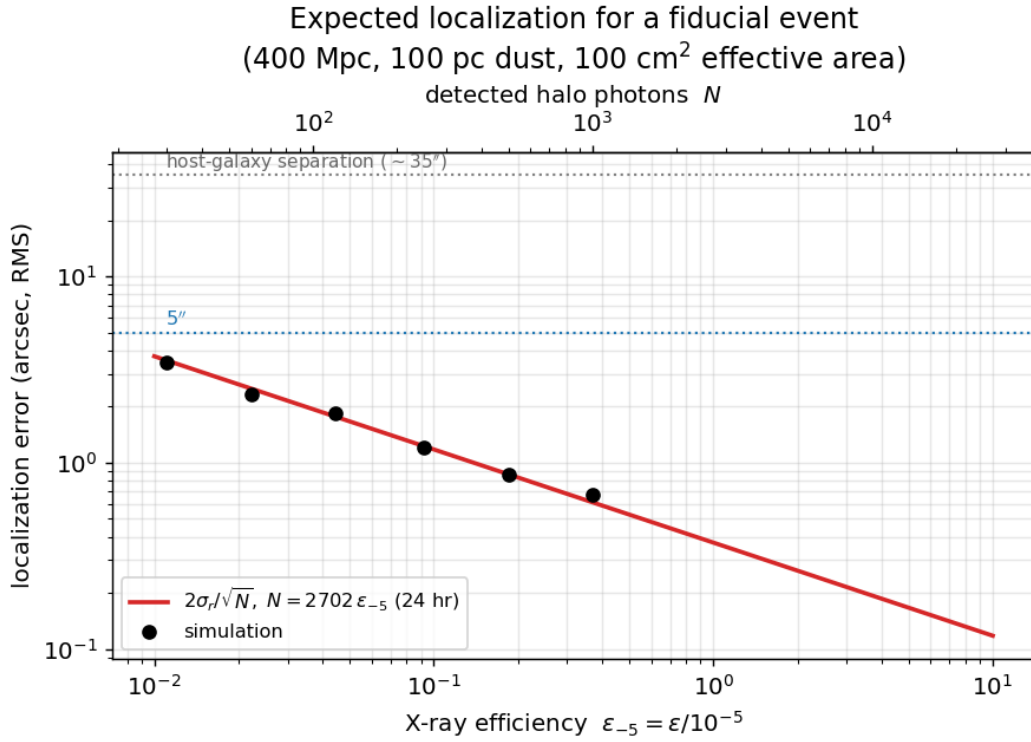


Figure 11: Expected localization error versus X-ray efficiency ϵ_{-5} for a fiducial event (400 Mpc, 100 pc dust path, 100 cm² effective area, so $N \approx 2700 \epsilon_{-5}$ collected within 24 hr), from the analytic law (red) and simulations (points). Dotted lines mark 5'' and the $\sim 35''$ host-galaxy separation. The top axis shows the corresponding photon count.

the scattering cross section, which sets how *bright* each ring is (rings at $\theta \gg \theta_0(E)$ are suppressed), not *where* it lies. For a set of discrete thin screens an energy-resolved image thus reweights the rings but does not move them. Only for spatially extended dust — where a range of distances contributes at a given delay — does the energy dependence of θ_0 reshape the radial *profile*, making higher-energy halos more compact; exploiting that is a possible refinement but is not the strong localization lever one might first assume.

The remaining idealizations are natural next steps, and not all of them are favorable. The dust is modeled as thin screens; a spatially extended cloud broadens its ring and inflates σ_r , degrading the Cramér–Rao bound $2\sigma_r/\sqrt{N}$ proportionally, with the uniformly distributed dust of NP20 as the unfavorable limit — real sightlines, in which a few discrete clouds dominate the column, lie between the two cases. A realistic calculation would draw column densities and distances from a three-dimensional Galactic dust map (e.g., Green et al. 2019) and build the ring template from data. The forward model also uses a single photon energy and a simplified Gaussian PSF; folding in a real instrument response, vignetting, and the source’s X-ray spectrum would make the precision estimates fully realistic. None of these is expected to change the central conclusion that a detected halo localizes its source to arcseconds.

A further idealization is that the screens are taken perpendicular to the line of sight. NP20 already treat the tilted case: for a screen inclined by α the halo remains circular, but its center is displaced by a physical distance $ct \tan \alpha$ — an angle $\delta = (ct/d) \tan \alpha \approx 2'' (t/1 \text{ day})(d/100 \text{ pc})^{-1}$ — and moves on the sky. The same result follows from the iso-delay condition used here: with $d(\theta, \phi) \simeq d_0(1 + \theta \tan \alpha \cos \phi)$, $\theta^2 d(\theta) = 2ct$ gives $\theta(\phi) \simeq \theta_{\text{ring}} - \frac{1}{2} \theta_{\text{ring}}^2 \tan \alpha \cos \phi$, a circle of unchanged radius displaced by

$$\delta = \frac{\theta_{\text{ring}}^2}{2} \tan \alpha = \frac{ct}{d} \tan \alpha.$$

What the present error budget adds is the comparison: at the fiducial epoch and distance $\delta = 0.43''$ for $\alpha = 45^\circ$ — between 12% and 65% of the statistical error over the range $N = 30\text{--}1000$ in Table 1 — so a tilt is sub-dominant throughout the regime considered here, though it would dominate for $N \gtrsim 2000$ or at late epochs ($1.7''$ at 24 hr). Two features make it tractable rather than merely limiting: the displacement grows *linearly* with t while the statistical error does not, so the drift NP20 note becomes a measurable signature rather than a nuisance; and screens at different distances are displaced by different amounts, so their rings cease to be strictly concentric — a second signature specific to the multi-screen halos considered here. Both are measurable with the multi-epoch imaging the search strategy of Section 4 already requires, which suggests that screen tilts could be fitted rather than assumed away.

6. Conclusion

Building on Nederlander & Paerels (2020), I have shown — with an explicit, statistically efficient estimator and a self-calibrating variant that needs no prior dust model — that a detected dust-scattering halo localizes its gravitational-wave source to about an arcsecond for a few hundred photons, saturating the fundamental Cramér–Rao limit, and does so even for a realistic multi-cloud sightline and without prior knowledge of the foreground dust. Because the same halo also measures the intervening dust distances and betrays itself through a $\sqrt{t}$ expansion, it offers not merely a delayed route to the sky position of a source whose prompt flash was missed, but — in most cases — to a unique host galaxy. The binding uncertainty is astrophysical, whether black-hole mergers emit detectable X-rays at all, rather than anything in the localization method.

Acknowledgments. This project was carried out independently and was motivated by the work of Nederlander & Paerels (2020); the author thanks Prof. Frits Paerels for introducing the topic. All simulations use only

NumPy and Matplotlib and are fully reproducible from the accompanying code.

Appendix A. Fiducial parameters, algorithm, and reproducibility

A.1. Fiducial parameters. Unless stated otherwise, the simulations use the values in Table A1, chosen to match the *Swift*/XRT-like case considered by NP20.

Table A1: Fiducial simulation parameters.

Parameter	Symbol	Value
Reference dust-screen distance	d	100 pc
Delay time of exposure	t_0	6 hr
Exposure duration	t_{exp}	1300 s
Photon energy	E	1 keV
Characteristic scattering angle	θ_0	10.4'
Grain-size range (scattering-weighted)	a	0.05–0.25 μm
PSF width (Gaussian dispersion)	—	0.15' (HPD $\approx$ 0.35')
Field of view	—	40' $\times$ 40'
Reference background surface density	—	0.07 arcmin $^{-2}$
Measured ring radial width	σ_r	9.7''
Fiducial source distance	D	400 Mpc
Fiducial effective area	A_{eff}	100 cm 2

A.2. Estimators. Two center estimators are used. The *template* estimator (Section 2.4) assumes the screen distances are known: it precomputes a PSF-smeared radial intensity template $s(r)$ from the ring geometry and maximizes $\sum_j \ln[(1-f)s(r_j) + fb]$ over the trial center by a coarse grid search plus two local-refinement stages. The background fraction f is supplied rather than fitted, at its expected value $B/(B+N)$ for the simulated exposure; this does not flatter the comparison with the Cramér–Rao bound of Section 2.5, because the Fisher information couples f to the center coordinates only through terms that vanish by azimuthal symmetry, so knowing f leaves the bound on (x_c, y_c) unchanged. The *self-calibrating* estimator (Section 3.5) assumes nothing about the dust: for each trial center it builds a smoothed radial density from the photons themselves (dividing out the $r dr$ area factor) and maximizes the summed log-density, so that the center is the point about which the radii cluster most sharply into rings. Ring radii for tomography (Section 3.6) are the peaks of the radial density about the recovered center, converted to distances by $d = 2ct/\theta^2$.

A.3. Reproducibility. The physics and estimators live in a single module (`halo.py`, NumPy only); separate scripts generate each figure and the Monte-Carlo results. No external data or network access is required, and every figure and quoted number in this paper is produced by running the accompanying code.

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